

An introduction to calculus

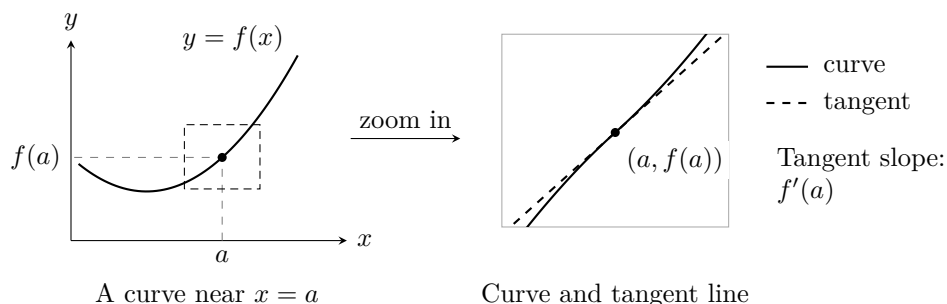
Math 1334

Wednesday, September 23, 2026

Before we begin: Syllabus, course outline, and questions.

1. What is calculus?

Calculus gives us a way to study change. In this course, we focus on *differential calculus*: how functions behave locally, very close to a point. The main idea is that a smooth curve looks nearly straight when we zoom in. The slope of that local line, the *tangent line*, is the derivative $f'(a)$.



For motion, this is the difference between *average velocity* over a time interval and *instantaneous velocity* at one moment. Limits will let us make the idea of “very close” precise.

2. How was calculus invented?

Newton and Leibniz independently developed calculus, building on earlier work. Their questions about tangents, areas, and motion became a common mathematical language. A few dates give the story some shape:

- **1664–1666.** Isaac Newton studies mathematics intensively at Cambridge, then develops the foundations of calculus during the plague years. The **1665** outbreak keeps him at Woolsthorpe for much of this period; he records basic calculus rules in **1666** but does not publish them immediately. [1]
- **1684.** Gottfried Wilhelm Leibniz publishes his first paper on differential calculus. Much of our notation, including dx and dy , comes from him. [2]
- **1684–1687.** Edmond Halley asks Newton about planetary motion in 1684. His questions and encouragement help lead to Newton’s *Principia* in **1687**; Halley also pays for its publication. [1]
- **1705 and 1758.** Halley uses Newton’s theory to predict a comet’s return in 1758. It returns as predicted. Halley’s comet last returned in **1986** and is expected again in **2061**. [3]

The point is that mathematics can do more than describe what we see: it can predict what happens next.

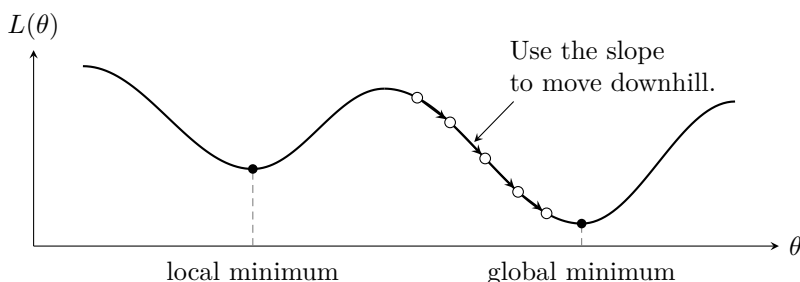
3. How do we use calculus?

Differential equations. These are equations involving rates of change. For an object of constant mass moving along a line, Newton’s law becomes

$$F = ma = m \frac{d^2x}{dt^2}.$$

Here $x(t)$ is position, its first derivative is velocity, and its second derivative is acceleration. A law about forces becomes a model of motion.

Optimization. Say we have a model with a parameter θ and a loss function $L(\theta)$ that measures how poorly it fits the data. We want to find a value of θ with small loss.



Idea: Use slopes of tangent lines to find extrema. How do we find a minimum without seeing the whole graph?

Gradient descent follows the downhill direction in small steps, an idea found in Cauchy’s **1847** paper. [4] It can settle in a local minimum and slow down where the graph is flat. Think of a ball moving down a landscape: the valley it reaches depends on where it starts.

Adam (adaptive moment estimation), introduced by Kingma and Ba in **2014**, uses running estimates of gradients and their squares to adjust the steps. [5] It connects this same slope idea to training machine learning models; it does not guarantee the lowest valley.

4. What will we learn about calculus?

- Weeks 1–3 Limits: making “very close” precise.
- Weeks 4–6 Derivatives: measuring local rates of change.
- Weeks 7–10 Applications: using derivatives to solve problems.
- Week 11 Final preparation: putting the ideas together.

References for the historical dates

- [1] The Newton Project, University of Oxford. *The Life and Work of Isaac Newton at a Glance*.
- [2] G. W. Leibniz (1684). *Nova methodus pro maximis & minimis*. *Acta eruditorum*, 467–473. HKUST Library digitization.
- [3] NASA Science. *1P/Halley*.
- [4] A.-L. Cauchy (1847). *Méthode générale pour la résolution des systèmes d’équations simultanées*. *Comptes rendus* 25, 536–538. English translation by R. J. Pulskamp.
- [5] D. P. Kingma and J. Ba (2014). *Adam: A Method for Stochastic Optimization*. arXiv:1412.6980; presented at ICLR 2015.